

MATH 200 - Elementary Linear Algebra

MATH 201 - Intermediate Linear Algebra

Section 6.1 Notes

Introduction to Linear Transformations

Devotional Thought

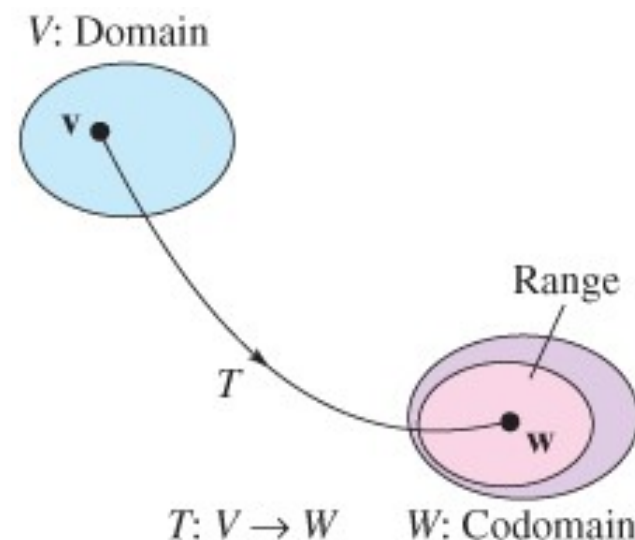
Images and Preimages of Functions

We now turn to the study of linear transformations, which are *functions* that **map** a vector space V into a vector space W . This type of function is denoted by

$$T : V \rightarrow W.$$

The standard function terminology is used for such functions.

For instance, V is the **domain** of T , and W is the **codomain** of T . If $\mathbf{v}$ is in V and $\mathbf{w}$ is in W such that $T(\mathbf{v}) = \mathbf{w}$, then $\mathbf{w}$ is the **image** of $\mathbf{v}$ under T . The set of all images of vectors in V is the **range** of T , and the set of all $\mathbf{v}$ in V such that $T(\mathbf{v}) = \mathbf{w}$ is the **preimage** of $\mathbf{w}$.



Remark There is a slight abuse of notation that will be used: For a vector

$$\mathbf{v} = (v_1, v_2, \dots, v_n)$$

in R^n , it would be more correct to use two sets of parentheses to denote $T(\mathbf{v})$ as $T(\mathbf{v}) = T((v_1, v_2, \dots, v_n))$. For convenience, we drop one set of parentheses to produce

$$T(\mathbf{v}) = T(v_1, v_2, \dots, v_n).$$

Example [Example 1] For any vector $\mathbf{v} = (v_1, v_2)$ in R^2 , define $T : R^2 \rightarrow R^2$ by

$$T(v_1, v_2) = (v_1 - v_2, v_1 + 2v_2).$$

(a) Find the image of $\mathbf{v} = (-1, 2)$.

(b) Find the image of $\mathbf{v} = (0, 0)$.

(c) Find the preimage of $\mathbf{w} = (-1, 11)$.

Linear Transformations

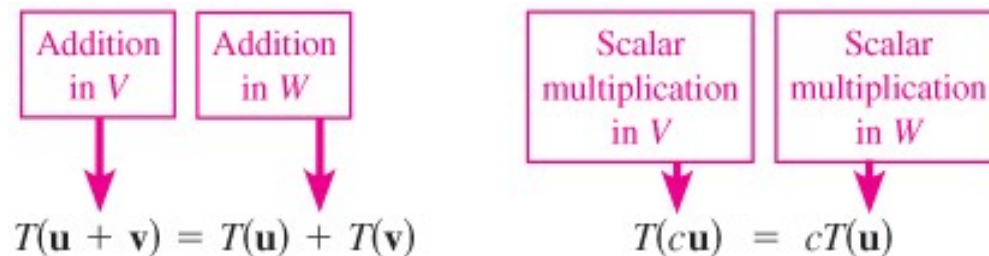
Our focus will be functions that map one vector space into another while preserving the operations of vector addition and scalar multiplication.

Definition Let V and W be vector spaces. The function $T : V \rightarrow W$ is a **linear transformation** of V into W when the two properties below are true for all $\mathbf{u}$ and $\mathbf{v}$ in V and for any scalar c .

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$

2. $T(c\mathbf{u}) = cT(\mathbf{u})$

A linear transformation is an example of a function that is *operation preserving* because the same result occurs whether you perform the operations of addition and scalar multiplication before or after applying the linear transformation. Although the same symbols denote the vector operations in both V and W , it should be noted that the operations must correspond to the vector space, as shown in the diagram below.



Example [Example 2] Show that the function in our first example is a linear transformation from R^2 into R^2 .

$$T(v_1, v_2) = (v_1 - v_2, v_1 + 2v_2)$$

Remark A linear transformation $T : V \rightarrow V$ from a vector space into itself (as above) is called a **linear operator** on V .

Example [Example 3] The following are *not* linear operators on R .

(a) $f(x) = \sin x$

(b) $f(x) = x^2$

(c) $f(x) = x + 1$

Remark The latter function suggests two uses of the term *linear*. The function $f(x) = x + 1$ is a linear *function* because its graph is a line. It *is not* a linear *transformation* from the vector space R into R , however, because it does not preserve vector addition or scalar multiplication.

Two simple linear transformations are the **zero transformation** and the **identity transformation**, which are defined below.

1. (Zero Transformation) $T : V \rightarrow W$ given by $T(\mathbf{v}) = \mathbf{0}$.

2. (Identity Transformation) $T : V \rightarrow V$ given by $T(\mathbf{v}) = \mathbf{v}$.

Properties of Linear Transformations

Note that the linear transformation in our first example had the property that the zero vector maps to itself. That is, $T(\mathbf{0}) = \mathbf{0}$. This property is true for all linear transformations, as stated in the first property of the theorem below.

Theorem 6.1 Let T be a linear transformation from V into W , where $\mathbf{u}$ and $\mathbf{v}$ are in V . Then the properties listed below are true.

1. $T(\mathbf{0}) = \mathbf{0}$
2. $T(-\mathbf{v}) = -T(\mathbf{v})$
3. $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v})$
4. If $\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n$, then

$$T(\mathbf{v}) = T(c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n) = c_1T(\mathbf{v}_1) + c_2T(\mathbf{v}_2) + \cdots + c_nT(\mathbf{v}_n).$$

The book proves properties 1, 2, and 3. Property 4 follows by considering that if T is a linear transformation, then it must preserve both vector addition and scalar multiplication, and thus T will have to preserve linear combinations as well.

Property 4 of Theorem 6.1 suggests that a linear transformation $T : V \rightarrow W$ is determined completely by its action on a basis for V . In other words, if $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a basis for the vector space V and if $T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_n)$ are given, then $T(\mathbf{v})$ can be determined for any $\mathbf{v}$ in V . (Put another way: T is *coordinate preserving*)

Example [Example 4] Let $T : R^3 \rightarrow R^3$ be a linear transformation such that

$$T(1, 0, 0) = (2, -1, 4),$$

$$T(0, 1, 0) = (1, 5, -2),$$

$$T(0, 0, 1) = (0, 3, 1).$$

Find $T(2, 3, -2)$.

Example [Exercise 30] Let $T : R^3 \rightarrow R^3$ be a linear transformation such that

$$T(1, 1, 1) = (2, 0, -1),$$

$$T(0, -1, 2) = (-3, 2, -1),$$

$$T(1, 0, 1) = (1, 1, 0).$$

Find $T(0, 2, -1)$.

Remark One further advantage of Theorem 6.1 is that it provides a quick way to identify functions that are not linear transformations. That is, all four conditions of the theorem must be true for a linear transformation, so it follows that if any one of the properties is not satisfied for a function T , then the function is not a linear transformation. For example, the function

$$T(x_1, x_2) = (x_1 + 1, x_2)$$

is not a linear transformation from R^2 into R^2 because $T(0, 0) \neq (0, 0)$.

Linear Transformations Given By Matrices

Example [Example 5] Define the function $T : R^2 \rightarrow R^3$ as

$$T(\mathbf{v}) = A\mathbf{v} = \begin{bmatrix} 3 & 0 \\ 2 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} \mathbf{v}_1 \\ \mathbf{v}_2 \end{bmatrix}.$$

(a) Find $T(\mathbf{v})$ where $\mathbf{v} = (2, -1)$.

(b) Show that T is a linear transformation from R^2 to R^3 .

Note that the latter argument would work for any such matrix example.

The following theorem says that multiplication by an $m \times n$ matrix is *always* a linear transformation from R^n into R^m .

Theorem 6.2 Let A be an $m \times n$ matrix. The function T defined by

$$T(\mathbf{v}) = A\mathbf{v}$$

is a linear transformation from R^n into R^m . In order to conform to matrix multiplication with an $m \times n$ matrix, $n \times 1$ matrices represent the vectors in R^n and $m \times 1$ matrices represent the vectors in R^m .

$$A\mathbf{v} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} a_{11}v_1 + a_{12}v_2 + \cdots + a_{1n}v_n \\ a_{21}v_1 + a_{22}v_2 + \cdots + a_{2n}v_n \\ \vdots \\ a_{m1}v_1 + a_{m2}v_2 + \cdots + a_{mn}v_n \end{bmatrix}.$$



Vector
in R^n



Vector
in R^m

Remark The $m \times n$ zero matrix corresponds to the zero transformation from R^n into R^m , and the identity matrix I_n corresponds to the identity transformation from R^n into R^n .

Class Exercise

Consider the linear transformation $T : R^n \rightarrow R^m$ defined by $T(\mathbf{v}) = A\mathbf{v}$. Find the dimensions of R^n and R^m for the linear transformation represents by each matrix.

$$(a) \ A = \begin{bmatrix} 0 & 1 & -1 \\ 2 & 3 & 0 \\ 4 & 2 & 1 \end{bmatrix}$$

$$(b) \ A = \begin{bmatrix} 2 & -3 \\ -5 & 0 \\ 0 & -2 \end{bmatrix}$$

$$(c) \ A = \begin{bmatrix} 1 & 0 & -1 & 2 \\ 3 & 1 & 0 & 0 \end{bmatrix}$$

Example [Exercise 44] Consider the linear transformation $T : R^5 \rightarrow R^3$ given by the matrix

$$A = \begin{bmatrix} 0 & 2 & 0 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 2 & 2 & 2 & 1 \end{bmatrix}$$

(a) Find $T(0, 1, 0, 1, 0)$.

(c) Find the preimage of $(1, -1, 2)$.